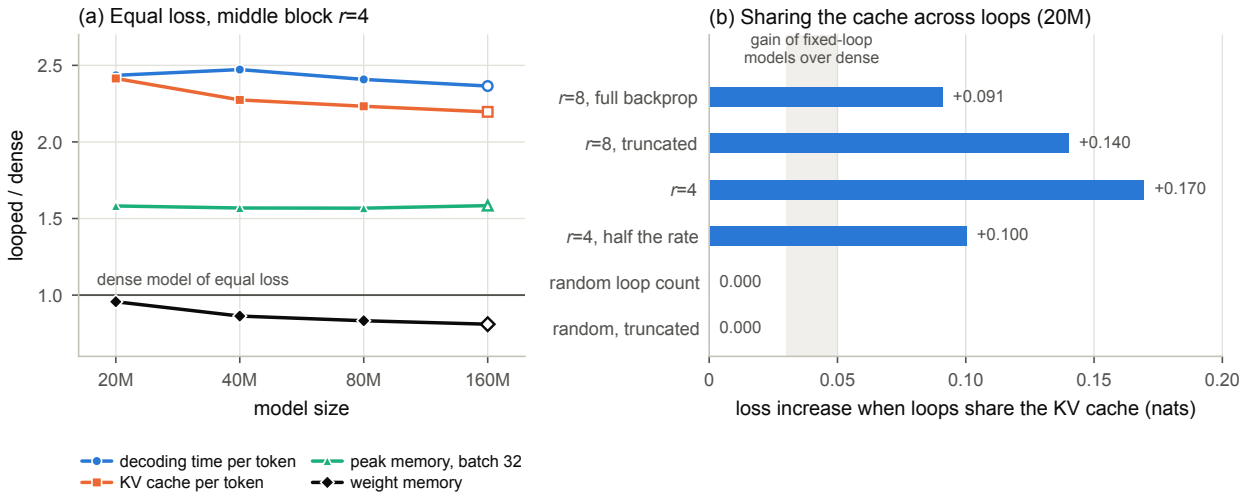
(a) Equal loss, middle block r=4
looped / dense
2.5
2.0
1.5
1.0
dense model of equal loss
20M
40M
80M
160M
model size
decoding time per token
KV cache per token
peak memory, batch 32
weight memory
(b) Sharing the cache across loops (20M)
gain of fixed-loop models over dense
r=8, full backprop
+0.091
r=8, truncated
+0.140
r=4
+0.170
r=4, half the rate
+0.100
random loop count
0.000
random, truncated
0.000
0
0.05
0.10
0.15
0.20
loss increase when loops share the KV cache (nats)